

Every positive finite Ermakoff limit occurs

A full answer to the footnote question in arXiv:2305.12873

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Status. Candidate full proof, likely valid, pending human review. For every prescribed $a \in (0, \infty)$ we construct a continuous strictly increasing function $g : [1, \infty) \rightarrow [1, \infty)$ with $g(1) = 1$ such that

$$\frac{tg(t)}{g(e^t)} = a$$

for all sufficiently large t . Thus the set of positive finite Ermakoff limits is the whole interval $(0, \infty)$.

1 Source question

Footnote 3 on source PDF page 5 recalls Ermakoff's test for a continuous positive increasing function $g : [1, \infty) \rightarrow [1, \infty)$ with $g(1) = 1$ and asks for an example in which

$$\lim_{t \rightarrow \infty} \frac{tg(t)}{g(e^t)}$$

exists but is different from 0, 1, and ∞ [1].

is an upper gradient of f_j . By the (X, Y) –Poincaré inequality [4] applied to each f_j on $2B$ we get

$$(6) \quad \|f_j - (f_j)_{2B}\|_{X(2B, \mu_{2B})} \leq c2r \|g\|_{Y(2\sigma B, \mu_{2\sigma B})}.$$

³V. P. Ermakoff [8] gave in 1872 a test for convergence of positives series based on the exponential function and the integral test. Namely, given a continuous, positive increasing function $g : [1, \infty) \rightarrow [1, \infty)$, such that $\lim_{t \rightarrow \infty} \frac{tg(t)}{g(e^t)} = a$, then the series $\sum_{n=1}^{\infty} \frac{1}{ng(n)}$ converges when $a < 1$ and diverges when $a > 1$ (a proof can be found in [18] p. 296 and p. 298).

We do not know any example where the limit exists but is different from 0, 1 or ∞ .

2 Proof intuition

Two logarithmic changes of variables turn the requested asymptotic ratio into an additive functional equation. Write

$$h(x) = \log g(e^x), \quad c = \log a.$$

For $t = e^x$ the identity $tg(t)/g(e^t) = a$ is equivalent to

$$h(e^x) = h(x) + x - c. \tag{1}$$

This equation can be solved one iterated-exponential interval at a time. Choose a short increasing seed that has exactly the endpoint jump required by (1); thereafter the recurrence itself propagates continuity and strict monotonicity forever.

3 Construction and proof

Theorem 1. *For every $a > 0$ there is a continuous strictly increasing $g : [1, \infty) \rightarrow [1, \infty)$ with $g(1) = 1$ for which*

$$\frac{tg(t)}{g(e^t)} = a \quad (2)$$

for every sufficiently large t .

Proof. Fix $a > 0$, put $c = \log a$, and choose

$$A > \max\{0, c\}, \quad E_0 = e^A, \quad m = \frac{A - c}{e^A - A} > 0.$$

Begin with the linear seed

$$h(x) = mx, \quad 0 \leq x \leq E_0. \quad (3)$$

Set $E_{n+1} = e^{E_n}$. Successively on each interval $[E_n, E_{n+1}]$, $n \geq 0$, define

$$h(y) = h(\log y) + \log y - c. \quad (4)$$

This is a genuine recursive definition: when $y \in [E_n, E_{n+1}]$, the argument $\log y$ lies in the part of the half-line on which h has already been defined. Equivalently, one may iterate logarithms until the argument first enters $[0, E_0]$ and then apply (3).

We verify the three properties needed below. At the first junction,

$$h(A) + A - c = mA + A - c = mA + m(e^A - A) = mE_0 = h(E_0),$$

so (4) agrees with the seed. At every later junction the two neighboring definitions use the same continuous expression in the already defined value $h(\log y)$. Induction therefore proves that h is continuous on $[0, \infty)$.

The seed is strictly increasing. If h is increasing on the previously constructed part and $y_1 < y_2$ lie in the next interval, then

$$h(y_2) - h(y_1) = h(\log y_2) - h(\log y_1) + \log y_2 - \log y_1 > 0.$$

Continuity at the junctions now proves by induction that h is strictly increasing everywhere. Finally, $h \geq 0$ on the seed interval, and each recursive step adds $\log y - c \geq A - c > 0$; hence h takes values in $[0, \infty)$.

For every $x \geq A$, the point e^x lies beyond E_0 , so the defining recurrence gives the exact identity

$$h(e^x) = h(x) + x - c. \quad (5)$$

Define

$$g(t) = \exp(h(\log t)), \quad t \geq 1. \quad (6)$$

The properties just proved show that g is continuous, strictly increasing, takes values in $[1, \infty)$, and satisfies $g(1) = 1$. If $t = e^x \geq e^A$, then (5) yields

$$\frac{tg(t)}{g(e^t)} = \exp(x + h(x) - h(e^x)) = \exp(c) = a.$$

This proves (2), in fact as an eventual exact identity rather than merely a limiting relation. $\square$

Corollary 2. *There is a continuous strictly increasing $g : [1, \infty) \rightarrow [1, \infty)$, $g(1) = 1$, for which the Ermakoff limit equals 2.*

Proof. In Theorem 1, take $a = 2$ and $A = 1$. Then

$$m = \frac{1 - \log 2}{e - 1}, \quad h(x) = mx \quad (0 \leq x \leq e),$$

and extend by $h(y) = h(\log y) + \log y - \log 2$ for $y \geq e$. The resulting g from (6) has $tg(t)/g(e^t) = 2$ for every $t \geq e$. $\square$

4 Checks, upgrade, novelty, and scope

The first construction used the explicit value $a = 2$, which already answers the source question. The deep upgrade in Theorem 1 shows that the same mechanism realizes every prescribed $a \in (0, \infty)$. The choice of a flexible threshold $A > \max\{0, \log a\}$ is what keeps the seed slope and every recursive increment positive, even when a is very large.

The script `code/verify_construction.py` evaluates the iterated-log formula, checks the functional equation and eventual ratios for several values of a , checks monotonicity on logarithmic grids, and probes both sides of the first iterated-exponential junctions. These numerical checks are not used in the proof.

A bounded primary-source search used the exact footnote wording, “Ermakoff’s condition”, $tg(t)/g(e^t)$, and the conjugated functional equation (1). It found the source question but no primary paper giving a nontrivial finite example or the arbitrary- a construction. Novelty confidence is moderate to high; correctness confidence is high.

The theorem answers only the footnote’s existence question. It does not alter the paper’s Poincaré inequality theorem, whose hypothesis specifically requires Ermakoff limit zero.

References

- [1] J. Martín and W. A. Ortiz, *A note on rearrangement Poincaré inequalities and the doubling condition*, arXiv:2305.12873.